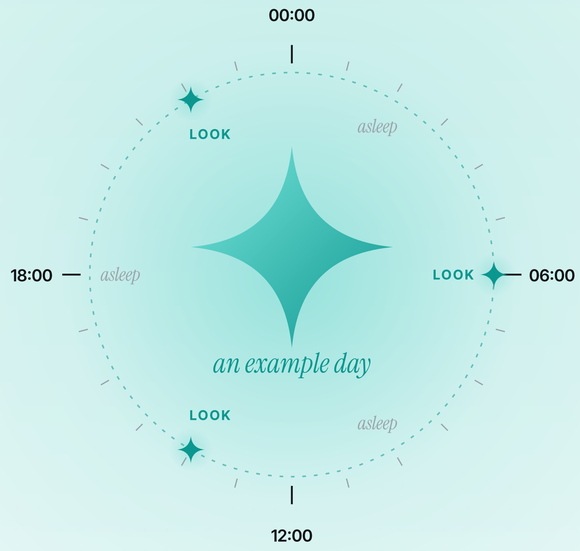
Maastricht University
BNAIC / BENELEARN 2026 · MAASTRICHT

A camera that *almost never* looks.

Tell it what matters in one sentence. A language model decides when a photo is worth taking.

Alexandru Cioc · Department of Advanced Computing Sciences, Maastricht University



illustrative

01

AN EXAMPLE DAY AT THE LOADING BAY

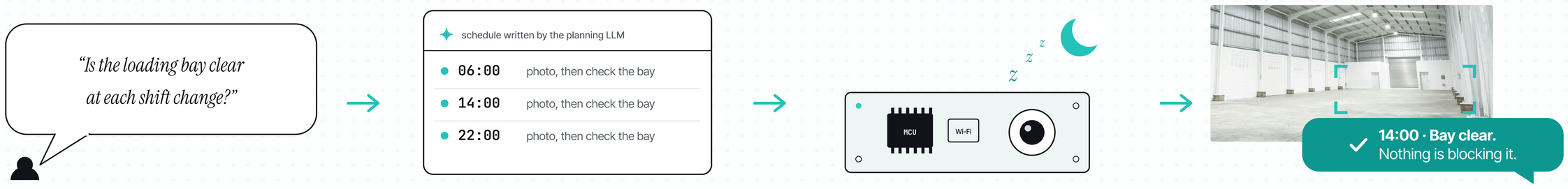
A motion sensor sees movement. *Hesperus sees the question.*



02

HOW IT WORKS

From a sentence to a photo, *and back to sleep.*



1 You ask

Describe what you want to know, in plain English.

2 A model plans

An LLM turns the goal into capture times and sends them to the board.

3 The camera sleeps

Any low-power camera board waits in deep sleep for each alarm. Prototype: a bare-metal STM32.

4 You get the answer

A vision model checks each photo against your goal and reports what it finds. Then the camera sleeps again.

03

WHAT IT COSTS

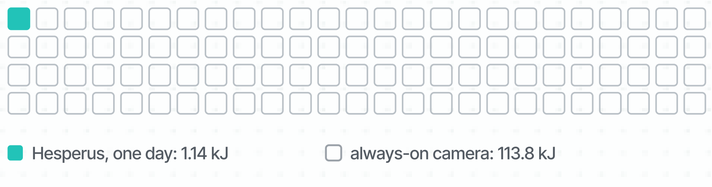
The camera sleeps. *The thinking happens elsewhere.*

95%



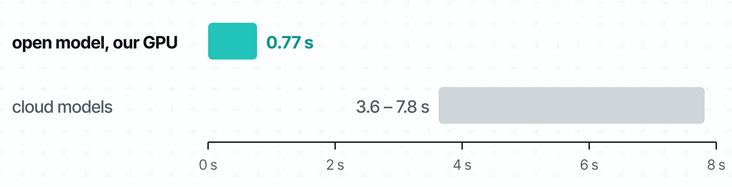
of **129 deployed hours** the board spent asleep

100×



less energy per day than an always-on camera (modelled)

0.8 s



response time per photo, open model on one GPU

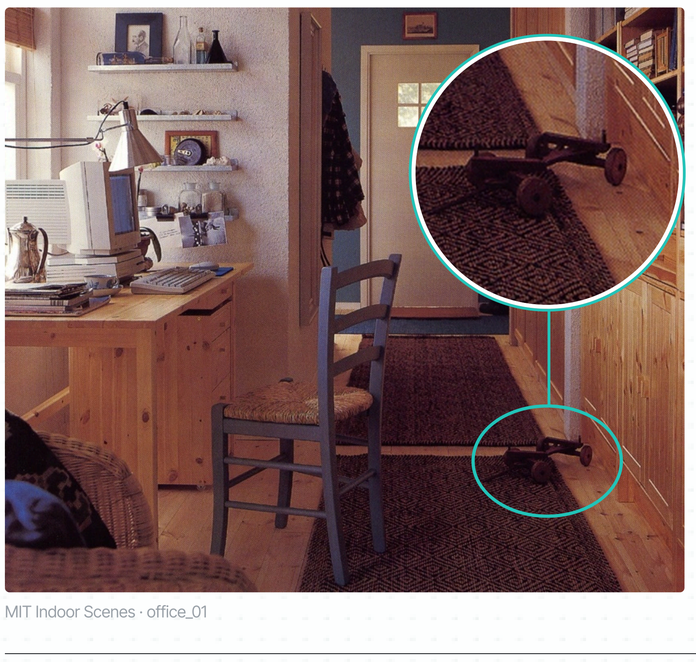
Energy is modelled from datasheet currents and checked against 129 hours of deployment logs; supply current was not metered. Response time: median, from sending a photo to receiving the written analysis.

04

SIX VISION MODELS, THE SAME PHOTO, TEN RUNS EACH

The eyes are swappable. *So we tested six.*

Goal: **"Monitor the workspace for activity and safety."** A wooden toy lies in the hallway, a trip hazard.



MIT Indoor Scenes · office_01

MODEL	QUALITY SCORE 0–9, see below	RESPONSE median	MENTIONED THE TOY in its answer, out of 10 runs
Gemini 3 Flash `gemini-3-flash-preview`	6.3	4.0 s	10 of 10
Claude Sonnet 4.6 `claude-sonnet-4-6 · thinking off`	6.6	7.0 s	0 of 10
Claude Sonnet 4.6 `claude-sonnet-4-6 · extended thinking`	6.2	7.8 s	0 of 10
Qwen3-VL 30B-A3B LOCAL `Q4_K_XL · llama.cpp · our GPU`	6.0	0.8 s	0 of 10
Claude Haiku 4.5 `claude-haiku-4-5-20251001`	5.9	3.6 s	0 of 10
Qwen2.5-VL 3B LOCAL `Q4_K_M · llama.cpp · our GPU`	4.7	0.5 s	1 of 10

Quality score: an LLM judge rates each answer 0–3 on finding hazards, describing the scene correctly, and not inventing things. The three add up to 0–9. Judge: Claude Sonnet 4.6; a second judge (Gemini) ranked the models the same way.

top five too close to call: the gaps could be chance (Kruskal–Wallis, p = 0.11) — likely range of each true score (95%)

Swap in a better model tomorrow. *The camera never notices.*

Today's models still miss hazards a person would catch. Because the thinking runs off the device, the next, better model drops in without touching the firmware or the battery.

Code on GitHub

Benchmark full write-up

Photos: Ryan Vargas / Unsplash; MIT Indoor Scenes